Assignment – 3

2. BMI Calculator

Code:

```
import java.util.Scanner;
class BMICalculator {
   double height;
   double weight;

   public BMICalculator() {
   }

   public BMICalculator(double height, double weight) {
      this.height = height;
      this.weight = weight;
   }

   public double getHeight() {
      return height;
   }
   public double getWeight() {
      return weight;
   }

   public void setHeight(double height) {
      this.height = height;
   }
   public void setWeight(double weight) {
      this.weight = weight;
   }

   public double calculateBMI() {
      return weight / (height * height);
   }
}

class BMITest {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.println("Enter height in m: ");
      double height = sc.nextDouble();
      System.out.println("Enter weight in kg: ");
      double weight = sc.nextDouble();

      BMICalculator bmiCalc = new BMICalculator();
      bmiCalc.setHeight(height);
      bmiCalc.setWeight(weight);

      double bmi = bmiCalc.calculateBMI();
      System.out.println("BMI: " + bmi);
   }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> javac BMITest.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> java BMITest
Enter height in m:
1.5
Enter weight in kg:
78
BMI: 34.666666666666664
PS D:\CDAC\PGCP-AI\assignment\Assignment -3>
```

4. Electricity

Code:

```
import java.util.Scanner;

class ElectricityBill {
   String customerName;
   double unitsConsumed;
   double billAmount;

   public ElectricityBill(String customerName, double unitsConsumed) {
      this.customerName = customerName;
      this.unitsConsumed = unitsConsumed;
   }

   public void calculateBillAmount() {
      if (unitsConsumed <= 100) {
         this.billAmount = this.unitsConsumed * 5;
      } else if (unitsConsumed <= 300) {
         this.billAmount = 100 * 5 + (unitsConsumed - 100) * 7;
      } else {
         this.billAmount = 100 * 5 + 200 * 7 + (unitsConsumed - 300) * 10;
      }
   }

   public void displayBill() {
      System.out.println("Customer Name: " + customerName);
      System.out.println("Units Consumed: " + unitsConsumed);
      System.out.println("Total Bill Amount: Rs. " + billAmount);
   }
}

class ElectricityCalc {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.println("Enter customer name: ");
      String Name = sc.next();
      System.out.println("Enter units consumed: ");
      double units = sc.nextDouble();

      ElectricityBill e = new ElectricityBill(Name, units);
      e.calculateBillAmount();
```

```
        e.displayBill();
    }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> javac ElectricityCalc.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> java ElectricityCalc
Enter customer name:
Priyansu
Enter units consumed:
3456
Customer Name: Priyansu
Units Consumed: 3456.0
Total Bill Amount: Rs. 33460.0
```

5. Telephone

Code:

```
import java.util.Scanner;

class TelephoneBill {

    String customerName;
    String phoneNumber;
    int numberOfCalls;
    float durationOfCalls;
    float billAmount;

    public TelephoneBill(String customerName, String phoneNumber, int numberOfCalls, float durationOfCalls)
    {
        this.customerName = customerName;
        this.phoneNumber = phoneNumber;
        this.numberOfCalls = numberOfCalls;
        this.durationOfCalls = durationOfCalls;
    }

    public void calculateBill(){

        float callCharge = 0;


        float minDuration = durationOfCalls;
        if (minDuration < 1) {
            minDuration = 1;
        }

        if (numberOfCalls <= 100) {
            callCharge = numberOfCalls * 0.50f;
        } else {
            callCharge = 100 * 0.50f + (numberOfCalls - 100) * 0.25f;
        }

        this.billAmount = 10 + callCharge;
```

```
    }

    public void displayBill(){
        System.out.println("Customer Name: " + customerName);
        System.out.println("Phone Number: " + phoneNumber);
        System.out.println("Number of Calls: " + numberOfCalls);
        System.out.println("Duration of Calls: " + durationOfCalls + " minutes");
        System.out.println("Total Bill Amount:" + billAmount);

    }
}

class TelephoneBillCalc {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter Customer Name: ");
        String name = sc.nextLine();
        System.out.println("Enter Phone Number: ");
        String phone = sc.nextLine();
        System.out.println("Enter Number of Calls Made: ");
        int calls = sc.nextInt();
        System.out.println("Enter Duration of Calls: ");
        float duration = sc.nextFloat();

        TelephoneBill teleBill = new TelephoneBill(name, phone, calls, duration);
        teleBill.calculateBill();
        teleBill.displayBill();

    }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> javac TelephoneBillCalc.ja
va
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> java TelephoneBillCalc
Enter Customer Name:
Priyansu
Enter Phone Number:
6372984535
Enter Number of Calls Made:
23
Enter Duration of Calls:
90
Customer Name: Priyansu
Phone Number: 6372984535
Number of Calls: 23
Duration of Calls: 90.0 minutes
Total Bill Amount:21.5
PS D:\CDAC\PGCP-AI\assignment\Assignment -3>
```

10. Credit Score Calculator

Code:

```
import java.util.Scanner;

class CreditScoreCalculator
{
   int creditHistory;
   double creditUtilization;

   public CreditScoreCalculator()
   {
      this.creditHistory = 0;
      this.creditUtilization = 0;
   }

   public CreditScoreCalculator(int creditHistory, double creditUtilization)
   {
      this.creditHistory = creditHistory;
      this.creditUtilization = creditUtilization;
   }

   public void setCreditHistory(int creditHistory)
   {
      this.creditHistory = creditHistory;
   }

   public void setCreditUtilization(double creditUtilization)
   {
      this.creditUtilization = creditUtilization;
   }


   public int getCreditHistory()
   {
      return this.creditHistory;
   }

   public double getCreditUtilization()
   {
      return this.creditUtilization;
   }


   public int calculateCreditScore()
   {
      int creditScore;

      if (this.creditUtilization < 30)
      {
         creditScore = (this.creditHistory * 15) + (int)(this.creditUtilization * 30) + 55;
      }
      else
      {
```

```
        creditScore = (this.creditHistory  * 15) + (int)(this.creditUtilization  * 30) + 35;
      }

      return creditScore;
    }
}

class CreditScoreCalc
{
   public static void main(String[] args)
   {
      Scanner sc = new Scanner(System.in);

      System.out.println("Enter credit history: ");
      int history = sc.nextInt();

      System.out.println("Enter credit utilization percentage: ");
      double utilization = sc.nextDouble();


      CreditScoreCalculator  csc = new CreditScoreCalculator(history,  utilization);

      int score = csc.calculateCreditScore();

      System.out.println("Calculated  Credit Score: " + score);
   }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> javac CreditScoreCalc.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> java CreditScoreCalc
Enter credit history:
4
Enter credit utilization percentage:
35
Calculated Credit Score: 1145
PS D:\CDAC\PGCP-AI\assignment\Assignment -3>
```

8. Rational Number;

Code:

```
import java.util.Scanner;


class RationalNumber
{
   int numerator;
   int denominator;


   public RationalNumber(int numerator, int denominator)
   {
      this.numerator = numerator;
      this.denominator = denominator;
   }


   public RationalNumber add(RationalNumber r)
   {
      int newNumerator = (this.numerator * r.denominator) + (r.numerator * this.denominator);
      int newDenominator = this.denominator * r.denominator;
      return simplify(newNumerator, newDenominator);
   }


   public RationalNumber subtract(RationalNumber r)
   {
      int newNumerator = (this.numerator * r.denominator) - (r.numerator * this.denominator);
      int newDenominator = this.denominator * r.denominator;
      return simplify(newNumerator, newDenominator);
   }


   public RationalNumber multiply(RationalNumber r)
   {
      int newNumerator = this.numerator * r.numerator;
      int newDenominator = this.denominator * r.denominator;
      return simplify(newNumerator, newDenominator);
```

```
}

public RationalNumber divide(RationalNumber r)
{
    int newNumerator = this.numerator * r.denominator;
    int newDenominator = this.denominator * r.numerator;
    return simplify(newNumerator, newDenominator);
}

public RationalNumber simplify(int num, int den)
{
    int smallest = num;
    if (den < num)
    {
        smallest = den;
    }

    int gcd = 1;
    for (int i = 1; i <= smallest; i++)
    {
        if (num % i == 0 && den % i == 0)
        {
            gcd = i;
        }
    }

    num = num / gcd;
    den = den / gcd;

    return new RationalNumber(num, den);
}

public void display()
{
```

```
        System.out.println(this.numerator  + "/" + this.denominator);
    }
}

class RationalNum
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the first rational number:");
        System.out.print("Numerator:  ");
        int num1 = sc.nextInt();
        System.out.print("Denominator:  ");
        int den1 = sc.nextInt();

        System.out.println("Enter the second rational number:");
        System.out.print("Numerator:  ");
        int num2 = sc.nextInt();
        System.out.print("Denominator:  ");
        int den2 = sc.nextInt();

        System.out.print("Enter the arithmetic operation  (+,-,*,/): ");
        String op = sc.next();

        RationalNumber r1 = new RationalNumber(num1,  den1);
        RationalNumber r2 = new RationalNumber(num2,  den2);
        RationalNumber result = new RationalNumber(0,  1);

        if (op.equals("+"))
        {
            result = r1.add(r2);
        }
        else if (op.equals("-"))
```

```
        {
            result = r1.subtract(r2);
        }
        else if (op.equals("*"))
        {
            result = r1.multiply(r2);
        }
        else if (op.equals("/"))
        {
            result = r1.divide(r2);
        }

        System.out.print("Result: ");
        result.display();
    }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> javac RationalNum.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> java RationalNum
Enter the first rational number:
Numerator: 23
Denominator: 46
Enter the second rational number:
Numerator: 23
Denominator: 454
Enter the arithmetic operation (+,-,*,/): +
Result: 125/227
PS D:\CDAC\PGCP-AI\assignment\Assignment -3>
```

3. Book Manager:

Code:

```
import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;

class Book {
    private String title;
    private String author;
    private String publisher;
    private String isbn;
    private int year;
    private double price;
    private int quantity;

    public Book() {
        this.title = "Unknown";
        this.author = "Unknown";
        this.publisher = "Unknown";
        this.isbn = "000-0000000000";
        this.year = 0;
        this.price = 0.0;
        this.quantity = 0;
    }

    public Book(String title, String author, String publisher, String isbn, int year, double price, int quantity) {
        this.title = title;
        this.author = author;
        this.publisher = publisher;
        this.isbn = isbn;
        this.year = year;
        this.price = price;
        this.quantity = quantity;
    }
```

```
public String getTitle() { return title; }
public String getAuthor() { return author; }
public String getPublisher() { return publisher; }
public String getIsbn() { return isbn; }
public int getYear() { return year; }
public double getPrice() { return price; }
public int getQuantity() { return quantity; }

public void setTitle(String title) { this.title = title; }
public void setAuthor(String author) { this.author = author; }
public void setPublisher(String publisher) { this.publisher = publisher; }
public void setIsbn(String isbn) { this.isbn = isbn; }
public void setYear(int year) { this.year = year; }
public void setPrice(double price) { this.price = price; }
public void setQuantity(int quantity) { this.quantity = quantity; }

public void increaseQuantity(int quantity) {
    if (quantity > 0) {
        this.quantity += quantity;
        System.out.println("Quantity increased. New quantity: " + this.quantity);
    } else {
        System.out.println("Increase amount must be greater than zero.");
    }
}

public void decreaseQuantity(int quantity) {
    if (quantity > 0 && quantity <= this.quantity) {
        this.quantity -= quantity;
        System.out.println("Quantity decreased. New quantity: " + this.quantity);
    } else if (quantity > this.quantity) {
        System.out.println("Not enough stock.");
    } else {
        System.out.println("Decrease amount must be greater than zero.");
```

```
        }
    }

    public double getInventoryValue() {
        return price * quantity;
    }

    public void displayBookInfo() {
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Publisher: " + publisher);
        System.out.println("ISBN: " + isbn);
        System.out.println("Year: " + year);
        System.out.println("Price: $" + price);
        System.out.println("Quantity: " + quantity);
        System.out.println("Inventory Value: $" + getInventoryValue());
    }
}


public class BookManager {
    private static List<Book> inventory = new ArrayList<>();
    private static Scanner scanner = new Scanner(System.in);

    public static void main(String[] args) {
        inventory.add(new Book("Clean Code", "Robert C. Martin", "Prentice Hall", "9734567865", 2008, 657, 10));
        inventory.add(new Book("The Pragmatic Programmer", "Andrew Hunt & David Thomas", "Addison-Wesley",
"974545555555", 1999, 498, 7));
        inventory.add(new Book("Introduction to Java", "Daniel Liang", "Pearson", "978454656446", 2021, 935, 15));

        int choice;
        do {
            printMenu();
            System.out.print("Enter choice: ");
            choice = scanner.nextInt();
```

```
        scanner.nextLine();
        if (choice == 1) viewAllBooks();
        else if (choice == 2) addBook();
        else if (choice == 3) increaseStock();
        else if (choice == 4) decreaseStock();
        else if (choice == 5) viewInventoryValue();
        else if (choice == 6) searchBook();
        else if (choice == 0) System.out.println("Goodbye!");
        else System.out.println("Invalid option. Try again.");
    } while (choice != 0);
    scanner.close();
}

private static void printMenu() {
    System.out.println("\nBook Inventory System");
    System.out.println("1. View All Books");
    System.out.println("2. Add New Book");
    System.out.println("3. Increase Book Quantity");
    System.out.println("4. Decrease Book Quantity");
    System.out.println("5. View Total Inventory Value");
    System.out.println("6. Search Book by Title");
    System.out.println("0. Exit");
}

private static void viewAllBooks() {
    if (inventory.isEmpty()) {
        System.out.println("No books in inventory.");
        return;
    }
    for (int i = 0; i < inventory.size(); i++) {
        System.out.println("Book #" + (i + 1));
        inventory.get(i).displayBookInfo();
    }
}
```

```
private static void addBook() {
    System.out.println("Add New Book");
    System.out.print("Title: "); String title = scanner.nextLine();
    System.out.print("Author: "); String author = scanner.nextLine();
    System.out.print("Publisher: "); String publisher = scanner.nextLine();
    System.out.print("ISBN: "); String isbn = scanner.nextLine();
    System.out.print("Year: "); int year = scanner.nextInt();
    System.out.print("Price: "); double price = scanner.nextDouble();
    System.out.print("Quantity: "); int quantity = scanner.nextInt();
    scanner.nextLine();
    Book newBook = new Book(title, author, publisher, isbn, year, price, quantity);
    inventory.add(newBook);
    System.out.println("Book added!");
    newBook.displayBookInfo();
}

private static void increaseStock() {
    System.out.println("Increase Book Quantity");
    Book book = selectBook();
    if (book == null) return;
    System.out.print("Units to add: ");
    int qty = scanner.nextInt();
    scanner.nextLine();
    book.increaseQuantity(qty);
}

private static void decreaseStock() {
    System.out.println("Decrease Book Quantity");
    Book book = selectBook();
    if (book == null) return;
    System.out.print("Units to remove: ");
    int qty = scanner.nextInt();
    scanner.nextLine();
```

```
    book.decreaseQuantity(qty);
}

private static void viewInventoryValue() {
    System.out.println("Inventory Value Report");
    double totalValue = 0;
    for (Book book : inventory) {
        double val = book.getInventoryValue();
        System.out.println(book.getTitle() + " - " + val + " (Qty: " + book.getQuantity() + ")");
        totalValue += val;
    }
    System.out.println("Total Inventory Value: " + totalValue);
}

private static void searchBook() {
    System.out.print("Enter title to search: ");
    String keyword = scanner.nextLine().toLowerCase();
    boolean found = false;
    for (Book book : inventory) {
        if (book.getTitle().toLowerCase().contains(keyword)) {
            book.displayBookInfo();
            found = true;
        }
    }
    if (!found) System.out.println("No book found with that title.");
}

private static Book selectBook() {
    if (inventory.isEmpty()) {
        System.out.println("No books available.");
        return null;
    }
    for (int i = 0; i < inventory.size(); i++) {
        System.out.println((i + 1) + ". " + inventory.get(i).getTitle());
```

```
        }
        System.out.print("Select book number: ");
        int index = scanner.nextInt() - 1;
        scanner.nextLine();
        if (index < 0 || index >= inventory.size()) {
            System.out.println("Invalid selection.");
            return null;
        }
        return inventory.get(index);
    }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> javac BookManager.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> java BookManager

Book Inventory System
1. View All Books
2. Add New Book
3. Increase Book Quantity
4. Decrease Book Quantity
5. View Total Inventory Value
6. Search Book by Title
0. Exit
Enter choice: 1
Book #1
Title: Clean Code
Author: Robert C. Martin
Publisher: Prentice Hall
ISBN: 9734567865
Year: 2008
Price: $657.0
Quantity: 10
Inventory Value: $6570.0
Book #2
Title: The Pragmatic Programmer
Author: Andrew Hunt & David Thomas
Publisher: Addison-Wesley
ISBN: 974545555555
Year: 1999
Price: $498.0
Quantity: 7
Inventory Value: $3486.0
Book #3
Title: Introduction to Java
Author: Daniel Liang
Publisher: Pearson
ISBN: 978454656446
Year: 2021
Price: $935.0
Quantity: 15
Inventory Value: $14025.0
```

6. Bank Manager:

Code:

```
import java.util.Scanner;

class BankAccount {

   String name;

   int accountNumber;

   double balance;


   BankAccount(String name, int accountNumber) {

      this.name = name;

      this.accountNumber = accountNumber;

      this.balance = 0;

   }

   void deposit(double amount) {

      this.balance = this.balance + amount;

      System.out.println("Deposit successful");

   }

   void withdraw(double amount) {

      if (amount <= this.balance) {

         this.balance = this.balance - amount;

         System.out.println("Withdrawal successful");

      }

      else {

         System.out.println("Insufficient balance");

      }

   }

   void displayBalance() {

      System.out.println("Current Balance: " + this.balance);

   }

   void displayInfo() {

      System.out.println("Account Holder Name: " + this.name);

      System.out.println("Account Number: " + this.accountNumber);

      System.out.println("Balance: " + this.balance);

   }
```

```
}
public class BankManager {
   public static void main(String[] args) {


      Scanner sc = new Scanner(System.in);
      BankAccount account = null;
      int nextAccountNumber = 1001;
      int choice;
      do {
         System.out.println("\n1. Create Account");
         System.out.println("2. Deposit");
         System.out.println("3. Withdraw");
         System.out.println("4. Check Balance");
         System.out.println("5. Display Account Info");
         System.out.println("6. Exit");
         System.out.print("Enter your choice:  ");
         choice = sc.nextInt();
         sc.nextLine();
         switch (choice) {
            case 1:
               System.out.print("Enter Account Holder Name: ");
               String name = sc.nextLine();
               account = new BankAccount(name, nextAccountNumber);
               System.out.println("Account created successfully");
               System.out.println("Your Account Number is: " + nextAccountNumber);
               nextAccountNumber++;
               break;


            case 2:
               if (account == null) {
                  System.out.println("No account found");
               }
      else {
                  System.out.print("Enter Account Number: ");
```

```
            int accNo = sc.nextInt();
            if (accNo == account.accountNumber) {
                System.out.print("Enter amount to deposit: ");
                double amount = sc.nextDouble();
                account.deposit(amount);
            } else {
                System.out.println("Invalid account number");
            }
        }
        break;
    case 3:
        if (account == null) {
            System.out.println("No account found");
        }
else {
            System.out.print("Enter Account Number: ");
            int accNo = sc.nextInt();
            if (accNo == account.accountNumber) {
                System.out.print("Enter amount to withdraw: ");
                double amount = sc.nextDouble();
                account.withdraw(amount);
            } else {
                System.out.println("Invalid account number");
            }
        }
        break;
    case 4:
        if (account == null) {
            System.out.println("No account found");
        } else {
            System.out.print("Enter Account Number: ");
            int accNo = sc.nextInt();
            if (accNo == account.accountNumber) {
                account.displayBalance();
```

```
                }
else {
                    System.out.println("Invalid account number");
                }
            }
            break;
        case 5:
            if (account == null) {
                System.out.println("No account found");
            }
    else {
                System.out.print("Enter Account Number: ");
                int accNo = sc.nextInt();
                if (accNo == account.accountNumber) {
                    account.displayInfo();
                } else {
                    System.out.println("Invalid account number");
                }
            }
            break;
        case 6:
            System.out.println("Thank you");
            break;
        default:
            System.out.println("Invalid choice");
        }
    } while (choice != 6);
  }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> javac BankManager.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> java BankManager

1. Create Account
2. Deposit
3. Withdraw
4. Check Balance
5. Display Account Info
6. Exit
Enter your choice: 1
Enter Account Holder Name: Priyansu
Account created successfully
Your Account Number is: 1001

1. Create Account
2. Deposit
3. Withdraw
4. Check Balance
5. Display Account Info
6. Exit
Enter your choice:
```

7. Toll Booth:
Code:

```
import java.util.Scanner;

class TollBooth {

   String vehicleType;

   int numAxles;

   double distanceTraveled;

   private double tollFee;

   private double totalAmountDue;

   void calculateTollFee() {

      if (vehicleType.equalsIgnoreCase("truck")) {

         tollFee = 0.50 * numAxles * distanceTraveled;

      }

      else {

         tollFee = 0.25 * numAxles * distanceTraveled;

      }

      totalAmountDue = tollFee + 2.00;

      System.out.println("Toll fee calculated successfully");

   }

   void generateBill() {

      if (tollFee == 0) {

         System.out.println("Please calculate toll fee first");

         return;

      }

      System.out.println("\n----- Toll Booth Bill -----");

      System.out.println("Vehicle Type: " + vehicleType);

      System.out.println("Number of Axles: " + numAxles);

      System.out.println("Distance Travelled: " + distanceTraveled + " miles");

      System.out.printf("Toll Fee: %.2f\n", tollFee);

      System.out.println("Processing Fee: 2.00");

      System.out.printf("Total Amount Due: %.2f\n", totalAmountDue);

   }

   static void showMenu() {

      System.out.println("\n1. Enter Vehicle Type (car, van, bus, truck)");
```

```
        System.out.println("2. Enter Number of Axles");
        System.out.println("3. Enter Distance Travelled");
        System.out.println("4. Calculate Toll Fee");
        System.out.println("5. Generate  Bill");
        System.out.println("6. Exit");
        System.out.print("Enter your choice:  ");
    }
}
public class Toll {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        TollBooth toll = new TollBooth();
        int choice;
        do {
            TollBooth.showMenu();
            choice = sc.nextInt();
            sc.nextLine();
            switch (choice) {
                case 1:
                    System.out.print("Enter vehicle type: ");
                    toll.vehicleType = sc.nextLine();
                    break;
                case 2:
                    System.out.print("Enter number of axles: ");
                    toll.numAxles = sc.nextInt();
                    break;

                case 3:
                    System.out.print("Enter distance travelled (in miles):  ");
                    toll.distanceTraveled = sc.nextDouble();
                    break;
                case 4:
                    toll.calculateTollFee();
                    break;
```

```
            case 5:
                toll.generateBill();
                break;
            case 6:
                System.out.println("Thank you");
                break;
            default:
                System.out.println("Invalid choice");
        }
    } while (choice != 6);
    sc.close();
  }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> javac Toll.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> java Toll

1. Enter Vehicle Type (car, van, bus, truck)
2. Enter Number of Axles
3. Enter Distance Travelled
4. Calculate Toll Fee
5. Generate Bill
6. Exit
Enter your choice: 1
Enter vehicle type: Car

1. Enter Vehicle Type (car, van, bus, truck)
2. Enter Number of Axles
3. Enter Distance Travelled
4. Calculate Toll Fee
5. Generate Bill
6. Exit
Enter your choice: 3
Enter distance travelled (in miles): 234

1. Enter Vehicle Type (car, van, bus, truck)
2. Enter Number of Axles
3. Enter Distance Travelled
4. Calculate Toll Fee
5. Generate Bill
6. Exit
Enter your choice:
```

9. Date:
code:

```
import java.util.Scanner;

class DateMethods  {

   int day;

   int month;

   int year;

   DateMethods(int  d, int m, int y) {

      day = d;

      month = m;

      year = y;

   }

   public boolean isLeapYear() {

      if (year % 400 == 0) {

         return true;

      } else if (year % 100 == 0) {

         return false;

      } else if (year % 4 == 0) {

         return true;

      } else {

         return false;

      }

   }

   public boolean isValid() {

      if (year < 1 || month < 1 || month > 12 || day < 1) {

         return false;

      }

      int[] days = {31, isLeapYear() ? 29 : 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

      if (day > days[month - 1]) {

         return false;

      }


      return true;

   }
```

```
public int getDayOfWeek() {
    int d = day;
    int m = month;
    int y = year;
    if (m < 3) {
        m = m + 12;
        y = y - 1;
    }
    int k = y % 100;
    int j = y / 100;
    int h = (d + (13 * (m + 1)) / 5 + k + (k / 4) + (j / 4) + (5 * j)) % 7;
    int result = (h + 6) % 7;
    return result;
}
public DateMethods getNextDay() {
    int[] days = {31, isLeapYear() ? 29 : 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
    int nd = day + 1;
    int nm = month;
    int ny = year;
    if (nd > days[month - 1]) {
        nd = 1;
        nm++;
        if (nm > 12) {
            nm = 1;
            ny++;
        }
    }

    return new DateMethods(nd, nm, ny);
}
public DateMethods getPreviousDay() {
    int nd = day - 1;
    int nm = month;
    int ny = year;
```

```
        if (nd < 1) {
            nm--;
            if (nm < 1) {
                nm = 12;
                ny--;
            }
            int[] days = {31, (ny % 400 == 0 || (ny % 4 == 0 && ny % 100 != 0)) ? 29 : 28,
                    31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
            nd = days[nm - 1];
        }
        return new DateMethods(nd, nm, ny);
    }
    public String toString() {
        String d = (day < 10 ? "0" + day : "" + day);
        String m = (month < 10 ? "0" + month : "" + month);
        return d + "-" + m + "-" + year;
    }
}
public class Date {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter day: ");
        int d = sc.nextInt();
        System.out.print("Enter month: ");
        int m = sc.nextInt();
        System.out.print("Enter year: ");
        int y = sc.nextInt();
        DateMethods date = new DateMethods(d, m, y);
        System.out.println("Is the date valid? " + date.isValid());
        System.out.println("Day of the week (0=Sunday ... 6=Saturday): " + date.getDayOfWeek());
        System.out.println("Is it a leap year? " + date.isLeapYear());
        DateMethods next = date.getNextDay();
        System.out.println("Next day is: " + next);
```

```
        DateMethods prev = date.getPreviousDay();

        System.out.println("Previous day is: " + prev);

    }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> javac Date.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -3> java Date
Enter day: 22
Enter month: 6
Enter year: 2000
Is the date valid? true
Day of the week (0=Sunday ... 6=Saturday): 4
Is it a leap year? true
Next day is: 23-06-2000
Previous day is: 21-06-2000
PS D:\CDAC\PGCP-AI\assignment\Assignment -3>
```